

Defining estimand for win ratio

How to separate true effect from censoring

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Outline

- **Background**
- **Censoring's impact on estimand**
 - Time frame of comparison
- **Two approaches to clarity**
 - **Nonparametric:** fix the time frame
 - **Semiparametric:** posit a time-constant feature
- **Summary and discussion**

Introduction

Composite Endpoint

- **Traditional composite endpoint (TCE)**

- **Time to first event**

- Major adverse cardiac event (MACE): death, heart failure, myocardio-infarction, stroke, etc.

- **Limitations:**

- Lack of clinical priority
 - Statistical inefficiency (waste of data)

- **Hierarchical composite endpoint (HCE)**

- **Example:** Death > nonfatal MACE > six-minute walk test (6MWT)/ NYHA class

Win Ratio: Basics

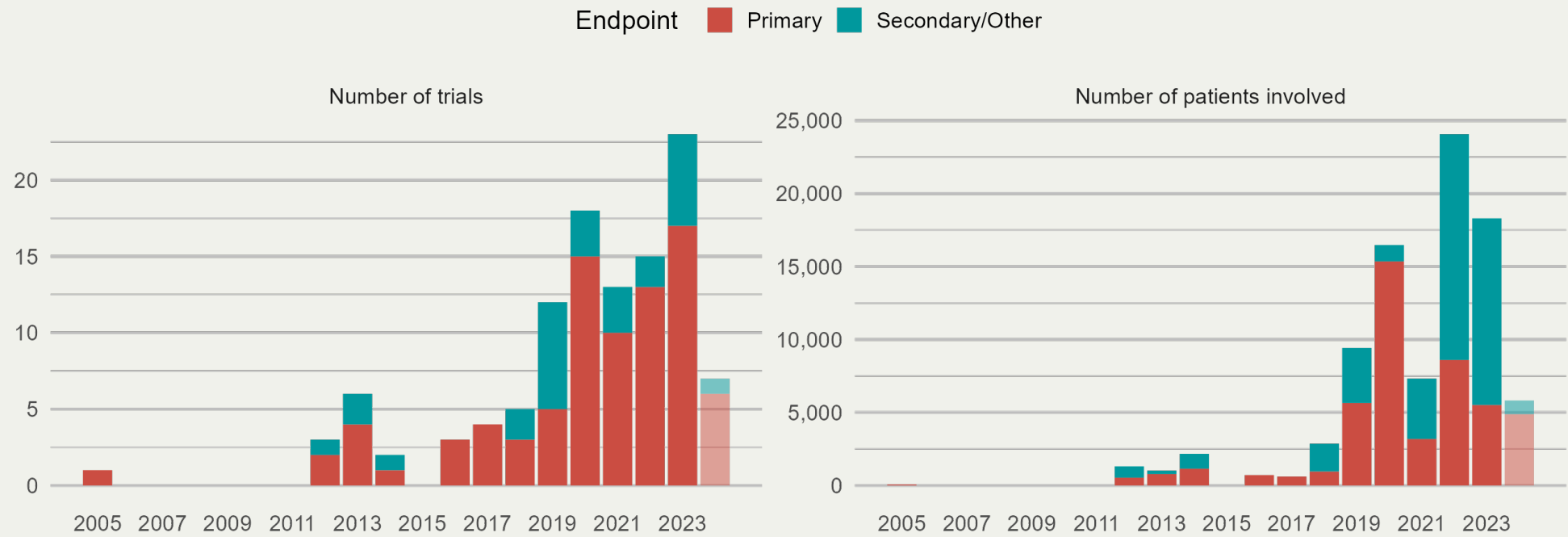
- **A common approach to HCE**
 - **Proposed and popularized** by Pocock et al. (2012)
 - **Treatment vs control:** generalized pairwise comparisons
 - **Win-loss:** sequential comparison on components
 - Longer survival > fewer/later nonfatal MACE > better 6MWT/NYHA score
 - **Effect size:** $WR = \text{wins} / \text{losses}$
- **Alternative metrics**
 - **Proportion in favor** (net benefit): $PIF = \text{wins} - \text{losses}$
 - **Win odds:** $WO = (\text{wins} + 2^{-1}\text{ties}) / (\text{losses} + 2^{-1}\text{ties})$

Win Ratio: Gaining Popularity

- More trials are using it...

Registered trials by start year

With win ratio-like approach to hierarchical composite endpoints



Source: ClinicalTrials.gov as of December 2023

Estimand vs Censoring

An Important Caveat

- **WR's estimand depends on censoring ...**
 - Luo et al. (2015), Bebu & Lachin (2016), Oakes (2016), Mao (2019), Dong et al. (2020a), Li et al. (2024), etc.
- **What is an estimand?**
 - Population-level quantity to be estimated
 - Population-mean difference, (true) risk ratio, etc.
 - Specifies how treatment effect is measured
 - **ICH E9 (R1) addendum:** estimand construction one of the “*central questions for drug development and licensing*” (ICH, 2020)

Time Frame of Comparison

- **Cause of dependency on censoring**

- Censoring → time frame of comparison → magnitude of win/loss probabilities

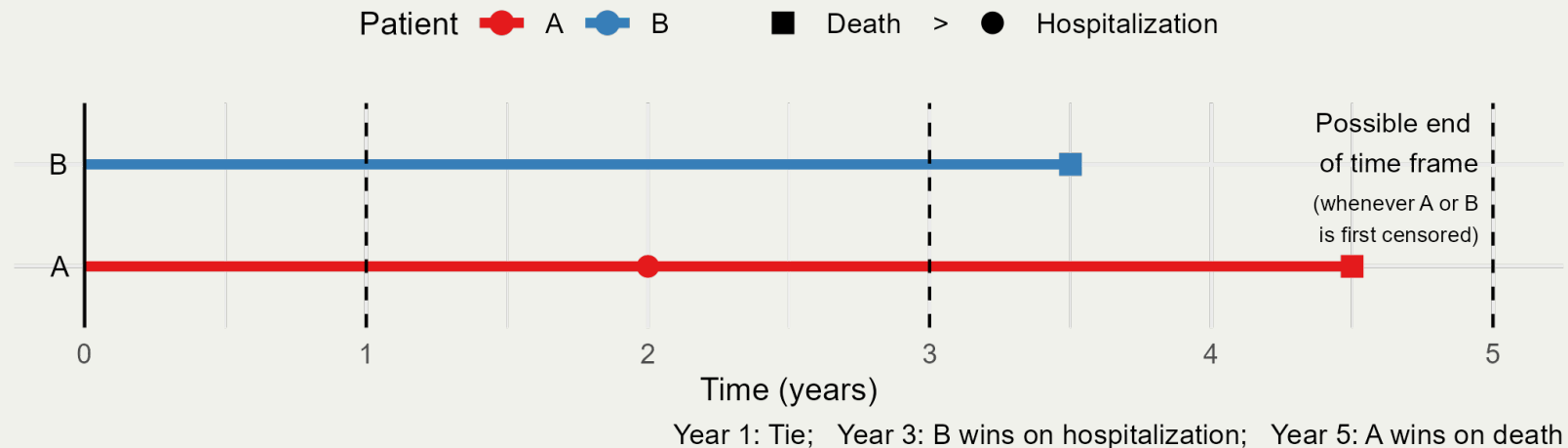
- **Example**

- **Pair 1:** one patient censored at year 1, the other > 1 year
 - Compared over $[0, 1]$ year
- **Pair 2:** neither patient censored until year 5
 - Compared over $> [0, 5]$ years
 - More events → fewer ties → higher win/loss proportions
 - Prioritized components more likely conclusive, harder to pass

Win-Loss Changes with Time

- Illustration

- Win-loss status, and deciding component, changes with time



- Longer follow-up ...

- **Parameters:** win/loss proportions $\uparrow$ (WR uncertain); tie proportion $\downarrow$
- **Relative contribution:** prioritized $\uparrow$; deprioritized $\downarrow$

Trial-Dependent Estimand

- **Actual estimand**

- Average WR mixing shorter-term with longer-term comparisons
- Weight set (haphazardly) by censoring distribution
 - Staggered entry, random withdrawal → non-scientific

- **Testing vs estimation**

- **Testing (qualitative):** okay
 - Valid under H_0 , powerful if treatment *consistently* outperforms control over time
- **Estimation (quantitative):** not okay
 - Needs a generalizable target quantity (scientific estimand)
 - Unaffected by length of trial, rates of patient accrual/loss to follow-up, etc.

Two Approaches to Meaningful Estimand

General Strategies

- **Goal:** a meaningful, generalizable WR estimand
 - Unaffected by censoring distribution
- **Key strategy**
 - Be *proactive* on time frame of comparison
- **Approaches**
 - Choose a fixed time (nonparametric)
 - Model time trajectory with a constant parameter (semiparametric)

Time Restriction - Univariate

- **Outcome data**

- $D^{(a)}$: survival time for a patient in group a (1: treatment; 0: control)
 - $S^{(a)}(t) = P(D^{(a)} > t)$

- **Time restriction: a familiar concept**

- *Five-year* survival rate of breast cancer patients
 - Estimand: $S^{(1)}(\tau) - S^{(0)}(\tau)$
- *Five-year* average survival time
 - Estimand: $E\{\min(D^{(1)}, \tau)\} - E\{\min(D^{(0)}, \tau)\}$
 - Restricted mean survival time (RMST)
- **Restriction time** $\tau = 5$ years (pre-specify)

Time Restriction - WR

- **Two-tiered composite**
 - $D^{(a)}$: survival time; $T^{(a)}$: time to (first) nonfatal MACE
- **Restricted win/loss probability**
 - Imagine all patients followed up to τ *[Math Processing Error]*
 - **Restricted WR:** $WR(\tau) = w_{1,0}(\tau)/w_{0,1}(\tau)$

Time Restriction - Estimation

- **General case**

- Formulate win/loss probability as function of time based on *uncensored* outcomes
- Pick restriction time τ

- **Estimation:** must handle censoring properly

- **Inverse probability censoring weighting** (IPCW, Dong et al., 2020b, 2021)
 - R-package: [WINS](#) (Cui & Huang, 2023)
- **Multiple imputations** for data censored before τ (T. Wang et al., 2023, 2024)

Time Restriction - A Variation

- **Take time difference into account**
 - $w_{a,1-a}(\tau)$: win probability by $\tau \rightarrow$ average win time by τ
 - Restricted mean time in favor: $w_{1,0}(\tau) - w_{0,1}(\tau)$ (Mao, 2023)
 - R-package: `rmt` (`rmt?`)
 - **Colon cancer trial**: levamisole + fluorouracil ($n = 304$) vs control ($n = 314$)

Table 1: Restricted mean times in favor of treatment in a colon cancer trial by $\tau = 7.5$ years.

	Estimate (yrs)	95% CI (yrs)	P-value
Death	0.62	(0.20, 1.04)	0.004
Recurrence	0.35	(0.21, 0.49)	<0.001
Overall	0.97	(0.47, 1.46)	<0.001

Temporal Modeling

- **Cox proportional hazards (PH) model**

- Time-varying hazard $\xrightarrow{\text{PH}}$ time-constant hazard ratio (global effect)
- Checking proportionality: score residuals

- **A proportional win-fractions (PW) model**

- Time-varying win-loss probability $\xrightarrow{\text{PW}}$ time-constant win ratio (global effect) (Mao & Wang, 2021; T. Wang & Mao, 2022)

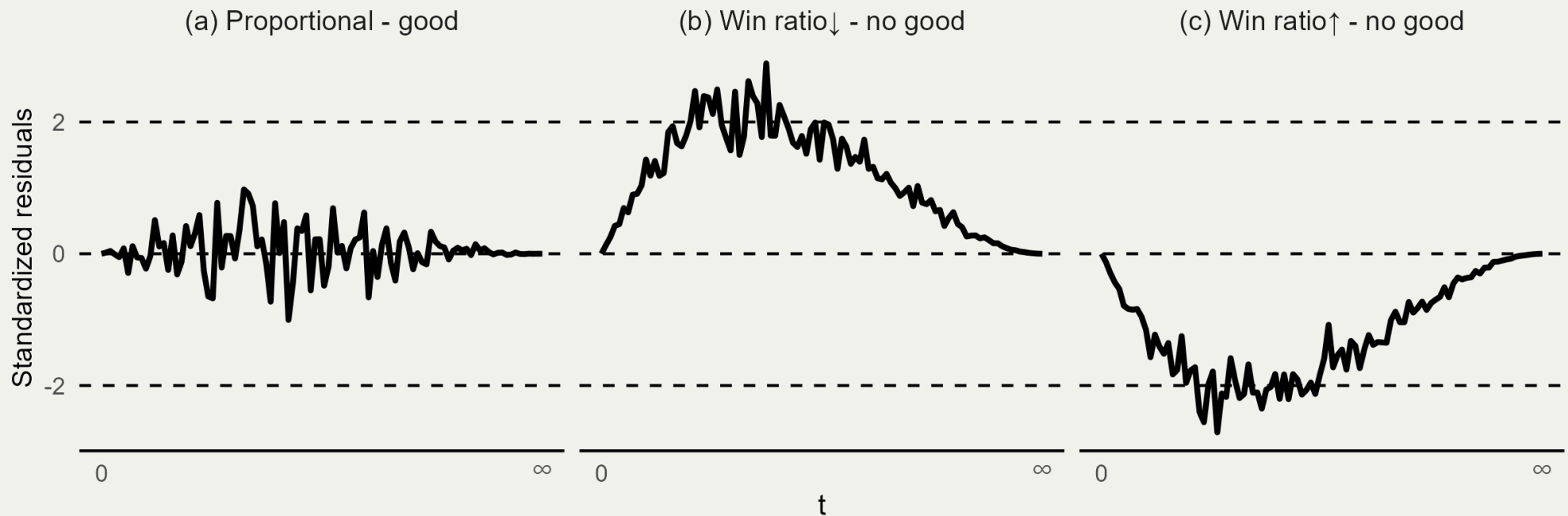
$$\frac{w_{1,0}(t)}{w_{0,1}(t)} = \exp(\theta) \text{ for some } \theta \text{ and all } t$$

- $\exp(\hat{\theta})$: standard or time-weighted WR statistic
- **R-package:** [WR](#) (WR?)

Checking Proportionality

- Cumulative residuals

$$\hat{\text{resid}}(t) = (\text{Observed wins by } t) - (\text{Model-based wins by } t)$$



Covariate Adjustment

- **HF-ACTION trial**

- **Exercise training** ($n = 1051$) vs **usual care** ($n = 1054$)

$$\frac{w_{z,z'}(t)}{w_{z',z}(t)} = \exp\{\theta(a - a') + \beta^T(x - x')\} \text{ for all } t$$

- **Covariates** x : sex, etiology, CPX, medical history, etc.

Table 2: Multiple PW regression for death > hospitalization in HF-ACTION.

	Win ratio	95% CI	P-value
Training v usual	1.06	(0.95, 1.19)	0.275
Male vs female	0.72	(0.63, 0.82)	<0.001
Ischemic vs non-ischemic	0.87	(0.76, 0.98)	0.027
CP exercise test (minute)	1.11	(1.09, 1.13)	<0.001
Atrial fibrillation	0.80	(0.70, 0.92)	0.002

Conclusion

Summary

- **How to separate true effect from censoring**
 - Make a conscious choice on time frame of comparison
 - Fix it (nonparametric) or model it (semiparametric)
- **Time restriction vs temporal modeling**
 - **Restricted win-loss:** model-free estimand, less efficient ([WINS](#), [rmt](#))
 - **PW regression:** may be more efficient *if proportionality (constant WR) holds* ([WR](#))
- **Combine the two**
 - IPCW + working model for locally efficient estimation?
 - Nonparametric estimand but semiparametric inference

Future Work

- **Sample size estimation**

- **Standard tests:** Gasparyan et al. (2021), Mao et al. (2022), Yu & Ganju (2022), B. Wang et al. (2023), etc.
- **Restricted WR:** ??

- **Intercurrent event**

- **Treatment non-response/toxicity/discontinuation** (ICH, 2020)
- **Hypothetical:** win/lose had treatment continued → imputation?
- **Composite:** death > *treatment failure* > lesser events?
- **Principal strata:** win/lose among those who would not experience treatment failure if assigned to either group (identifiability)

For More

- **Main paper**

Mao, L. (2024). Defining estimand for the win ratio: separate the true effect from censoring. *Clinical Trials*, 21, 584-594. <https://doi.org/10.1177/17407745241259356>

- **Slides URL:** https://lmaowisc.github.io/ce/win_ratio_estimand.html

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